MeridianHealth

Retail & Supply Chain

End-to-End Data Analysis Project

THE QUESTION YOU ARE ANSWERING

"How can operational data from 12 pharmacy stores drive decisions that grow revenue, cut inventory waste, optimise staffing, and deepen customer loyalty?"

TOOLS	DATASET	PERIOD	DELIVERABLES
Python · Databricks · Power BI	9 Tables · 35,000+ Rows	Sep 2026	Cleaned Data · Dashboard · Report

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BUSINESS CONTEXT

1 Why This Project Exists

The Business Context

MeridianHealth is a pharmacy retail group operating 12 stores across Canada with 120 employees and approximately 1,500 registered customers. Operations generates data across sales, prescriptions, inventory, staff and suppliers — but it has never been cleaned, validated, or analysed in a structured way.

Five Business Questions Management Needs Answered:

1. Which stores are genuinely profitable after cost and discount adjustments?
2. Which product categories drive revenue — and which are over-discounted?
3. How much inventory value is at expiry risk right now?
4. Is pharmacist staffing aligned with actual prescription workload?
5. Which customer segments and payment channels generate the most value?

THE DATA

2 What You Are Working With

TABLE NAME	DESCRIPTION	ROWS
fact_sales	One row per sale (store, product, employee, customer)	~9,900
fact_prescriptions	Prescription records linked to customer, product, doctor	~10,000
fact_stock	Inventory snapshot: quantity, expiry, reorder flags	~10,000
dim_customer	Registered customer profiles with loyalty tier & city	~1,500
dim_product	Product catalogue with cost price, category, supplier	~5,000
dim_employee	Staff: position, pharmacist flag, store assignment	120
dim_store	Store locations, regions and manager names	12
dim_category	Product category reference (9 categories)	9
dim_supplier	Supplier: city, contact person, phone	5

Schema type: Galaxy-Snowflake — 3 fact tables share conformed dimensions; dim_supplier branches off dim_product (snowflake arm).

PHASE 1

3 Data Cleaning in Python

Objective

Load all 9 CSV files using pandas, inspect and fix each table, then export clean versions to a /cleaned/ folder ready for Databricks ingestion.

Cleaning Tasks You Must Complete

- Inspect dtypes for all 9 tables — cast dates to datetime, IDs to int/str, numerics to float.
- Handle null values: document null counts per column; decide fill-or-drop per business logic.
- Remove exact duplicate rows; flag near-duplicates in fact tables by key column combinations.
- Standardise text: strip whitespace, consistent casing (Title Case for names, UPPER for codes).
- Validate value ranges: negative quantities, discount_pct > 1.0, margin_pct outliers beyond ±3σ.
- Export 9 cleaned CSVs to /cleaned/ sub-folder — naming pattern: clean_<table_name>.csv.

Key Questions to Answer in Your Cleaning Notebook

- What is the total null percentage across all 9 tables? Which table is worst?
- Are there product IDs in fact_sales that do not exist in dim_product?
- How many employees have a position/is_pharmacist flag mismatch (like employee 97)?
- What is the margin_pct distribution in fact_sales — are any values suspicious?

PHASE 2

4 Validation Script

Cross-Table Integrity Checks

- FK check: every fact_sales[product_id] must exist in dim_product[product_id] — log orphan count.
- FK check: every fact_sales[store_id] must exist in dim_store[store_id].
- FK check: every fact_prescriptions[customer_id] must exist in dim_customer[customer_id].
- Assert join success rate > 99 % for all FK pairs; fail the script if any pair falls below.
- Output a referential integrity report: table pair · total rows · match % · orphan count.

Business Rule Validation

- Pharmacist flag mismatch: flag employees where position='Pharmacist' AND is_pharmacist=0, or vice-versa.
- Expiry logic: flag fact_stock rows where expiry_date < today AND quantity > 0 (expired, still in stock).
- Margin sanity: flag fact_sales where margin_pct < 0 (below cost) or margin_pct > 0.80 (above 80 %).
- Discount cap: flag rows where discount_pct > 0.50 (above the 50 % policy ceiling).
- Stock flag inconsistency: flag where quantity ≤ reorder_level but below_reorder_flag = 0.
- Script output: export validation_report.csv — row_id, table, rule_violated, severity (WARN / ERROR).

A passing validation script is a prerequisite before uploading data to Databricks.

PHASE 3

5 Databricks Lakehouse

Upload & Delta Tables

■ Upload 9 cleaned CSVs to DBFS via Files > Upload.
■ CREATE SCHEMA meridianhealth in Databricks.
■ Create 9 Delta Tables: CREATE TABLE ... USING DELTA.
■ Run DESCRIBE HISTORY to verify Delta log is active.
■ Demo Time Travel: SELECT ... VERSION AS OF 0 vs current.

SQL Queries on Delta

■ Top 5 stores by total revenue (GROUP BY store_id DESC).
■ Average margin % per category (JOIN dim_category).
■ Expiry-risk value: SUM(qty * cost_price) WHERE flag=1.
■ Pharmacist count vs prescriptions per store — joined query.
■ Revenue % by payment method across all stores.

Lakeview Dashboard (bonus) Build a 3-chart Databricks dashboard before switching to Power BI.
Charts: Revenue by Store · Expiry Risk by Category · Prescriptions per Pharmacist

PHASE 4

6 Power BI Report

Data Model (Galaxy-Snowflake)

■ Import cleaned CSVs — Import Mode
■ fact_sales → dim_product, dim_store, dim_employee, dim_customer (*:1).
■ fact_stock → dim_product (*:1).
■ fact_prescriptions → dim_product (*:1).
■ dim_supplier → dim_product (*:1) — the snowflake arm.
■ Fix many-to-many auto-detections; verify all relationships are active.
■ Use LOOKUPVALUE() as fallback if RELATED() raises a context error.

Required DAX Measures

■ Total Revenue = SUM(fact_sales[total_amount])
■ Weighted Margin % = DIVIDE(SUM(profit), SUM(total_amount))
■ Expiry Risk Value = SUMX(fact_stock, RELATED(cost_price)) WHERE flag=1
■ Prescriptions per Pharmacist = DIVIDE([Rx Count], [Pharmacist Count])
■ is_pharmacist Mismatch Count — dual-condition CALCULATE
■ Repeat Customer Revenue % — DIVIDE with loyalty segment filter

Report Pages — Design Rules

■ Page 1 — Store & Product: Revenue by store (bar), Margin % by category (bar), Discount rate vs revenue scatter, KPI cards for Top/Bottom 3 stores.
■ Page 2 — Inventory, Staffing & Customers: Expiry risk by category (bar), Prescriptions per pharmacist (bar), Customer segment revenue % (donut), Payment method split (bar).
■ Design rule: every page answers ONE question stated in its subtitle. Use the HTML Content visual for navy-to-teal gradient KPI cards.

PHASE 5 — FINAL DELIVERABLE

7 Insights & Strategic Recommendations

Key Findings from the Data — 5 FINDINGS

1. Store Profitability Gap

Margin % varies significantly across the 12 stores. At least two consistently operate below 5 % weighted margin — likely driven by above-average discount rates or shrinkage. These stores require immediate review before the next planning cycle.

2. Over-Discounting in Two Categories

Vitamins & Supplements and Baby Care carry the highest average discount rates while delivering below-average returns. Revenue share does not justify the discount depth; both categories take more from the P&L than they return.

3. Material Inventory Value at Expiry Risk

fact_stock shows a significant volume of near-expiry units (expiry_risk_flag=1) across Pharmaceuticals and Vitamins. Holding to expiry converts an asset into a 100 % write-off. Clearance promotion or supplier return must be activated now.

4. Pharmacist Staffing Misalignment

Prescription volume per pharmacist varies 3× across stores, and the is_pharmacist flag has data-quality mismatches. The HR team must correct records before any staffing decision — the data cannot yet be trusted for headcount planning.

5. Loyalty Customers Punch Above Their Weight

Gold-tier loyalty customers represent a small share of total customers but generate a disproportionate share of repeat-purchase revenue. Card/digital payment customers show higher average order values — a signal to invest in the loyalty programme.

Strategic Recommendations — 3 ACTIONS

R1 Activate clearance for all expiry-flagged stock within 30 days. Any unit sold at 40 % discount is preferable to a 100 % write-off. Prioritise Pharmaceuticals — they carry the highest per-unit cost_price.

R2 Reduce the discount authorisation ceiling to 20 % for Vitamins & Baby Care, pending a formal promotion ROI review. Every 1 % reduction in discount rate on these categories directly improves chain-wide weighted margin.

R3 Commission a data-quality sprint on dim_employee before any staffing decision. Correct all is_pharmacist flag mismatches, then re-run the Prescriptions-per-Pharmacist analysis to identify which stores are genuinely under- or over-staffed.

All findings are based on training data. Numbers are designed for relative comparison — do not quote as real-world benchmarks.